

Reply to S. Danilov’s comments on the mEVP communication study

N. Koldunov, 6 August 2026

We built the variant you describe, and it changes the study’s conclusion. Written as you write it — the ring computation folded into the loops that already exist rather than into kernels of its own — the wide halo removes **57 to 59% of the sea-ice cost** on CORE2 at its operating point and **37 to 43%** on fArc, against 11% and 6 to 11% for our version of the same scheme. **The rewriting multiplies the saving by three to seven.** The paired runs compute the same ring and put identical message and byte counts on the wire; they differ by 360 kernel launches per time step. Our earlier conclusion, that exactness costs most of the saving, was measuring our implementation.

each mesh, six legs in one allocation	CORE2, 1 node / 4 GPU’s		fArc, 4 nodes / 16 GPU’s	
	step	ice cost	step	ice cost
every sub-cycle (reference)	0.0667 s	15.7 ms	0.1162 s	26.0 ms
<i>delayed exchange (stale halo), $K=8$</i>	−14.4%	−58.0%	−13.7%	−62.7%
wide halo, standard form, our kernels	−3.0%	−11.5%	−2.2%	−10.8%
wide halo, divergence form, our kernels	+1.1%	+2.6%	−1.0%	−5.8%
wide halo, standard form, your loops	−14.1%	−56.7%	−7.8%	−36.5%
wide halo, divergence form, your loops	−14.4%	−58.6%	−9.6%	−43.1%

The delayed exchange is in the table because it is the scheme we rejected for putting the domain decomposition into the ice field, and because we reported it as saving about three times what the wide halo saves. That factor was our kernel structure. On CORE2 it is gone entirely — the divergence form matches the delayed exchange on the step and is 1.5% ahead of it on the ice. On fArc it is not: the rewriting closes half to two thirds of the gap and the delayed exchange still leads. We would not now describe the wide halo as a fraction of it, but neither can we claim parity in general on two meshes.

Giving up the owner’s summation order costs what you predicted: after a time step of 120 sub-cycles the ice velocity differs from the exact wide halo by 8 to 15 units in the last place. The 60-day test of the domain decomposition, which replaces the bit-identity we gave up, passes: the ratio of boundary to interior differences is 0.84 for ice thickness against 0.85 for two identical runs, and on ice velocity 1.20 against a control of 1.33, while the delayed exchange gives 2.5 and 1.76.

Point 1, the auxiliary fields. We instrumented the per-step exchange to report, field by field, the difference between the owner’s arriving value and the value the receiving process already holds. Nine of the eleven arrive *exactly* equal — ice velocity, concentration, thickness, snow, ocean velocity, wind stress — because FESOM’s own halo exchange has already put them there, as you said. The two right-hand sides do not: they are assembled and area-scaled over owned nodes only, so a halo node has no correct value at all. But the redundancy stops at the first ring, and rings $2 \dots K$ have no other source, so what could be removed is 37% of that exchange at $K=2$ and 17% at $K=4$. Since the exchange is 5.4% of the data the scheme moves and one message per neighbour per step, removing it would gain under one percent of the traffic and no messages. We have not implemented the trimmed version, and that is why.

Point 1, the kernel structure. This is the part we could not bound before, and it was much the larger. Our wide halo adds 362 kernel launches per model step, each on a slower launch path than the owned kernel it duplicates: the ring kernels carry enough array descriptors to cross a threshold in Kokkos above which arguments are staged through constant memory. The gap between consecutive launches is $14.3 \mu\text{s}$ for those against $4.4 \mu\text{s}$ for the owned kernels. Per step the ring kernels cost 5.9 ms of inter-kernel idle against 2.5 ms of arithmetic.

Your reformulation now wins, and we had been measuring past it. With ring kernels the standard form beat the divergence form by 14 points of ice cost on CORE2 and 5 on fArc, and our previous note decomposed that gap into a message term and a byte term. With the ring folded into the existing loops *the ordering reverses*: the divergence form leads by 1.9 points on CORE2 and 6.5 on fArc, growing with the mesh, which is the behaviour the halved nodal message should produce. The launch structure was not creating the difference between the two forms, it was hiding it. On CPU the same rewriting also favours your form — there the ring assembly is arithmetic-bound, and folding the loops removes a recomputation the divergence form was doing about six times over, worth 3 to 5 percentage points, while in the standard form it costs 6. Neither is profitable on CPU in either writing, at 512 or 2048 cores.

Point 2, extending the computation over the ring. You were right, and the correction runs deeper than we allowed. Recomputing the ring does remove the imprint — averaging $|\Delta \text{ ice thickness}|$ on the sub-domain boundaries against the interior over 60 days on fArc gives 0.87, against 0.85 for two identical runs and 2.52 for the delayed exchange. What we wrote was that the same change removes most of the saving and that we had found no intermediate that was both cheap and clean. That was a statement about our kernels, not about the scheme.

Open. DARS and NG5 are still queued. Both are points where each GPU holds more of the mesh than on fArc, so they are where the divergence form’s lead should be largest if the trend above is real.

All numbers are from the runs described in `danilov_mevp_report.pdf`; the output and job scripts can be shared.